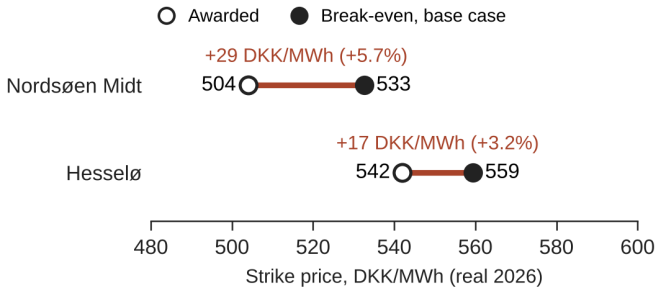
Reverse-engineering Vattenfall's 2026 Danish offshore wind bids

What do DKK 504/MWh (Nordsøen Midt) and DKK 542/MWh (Hesselø) imply about cost, return and wind?

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Vattenfall's winning prices are 3–6 % below what a public-data model needs to break even. They work at a real after-tax return of about 4.5–4.7 %, about 1 pp above Vattenfall's own discount rate for wind, or if it builds 4–6 % cheaper than my estimate from Danish Energy Agency (DEA) data. With 15 MW turbine costs the return falls below that rate: the bids look like a bet on 20 MW turbines. Nordsøen Midt carries more downside: its cap on state payments can run out if power prices are low.

The gap: awarded price vs my base-case break-even



Any one of these closes the gap

	Nordsøen Midt	Hesselø
Return, real after tax	4.5 %	4.7 %
Build cost, DKK m/MW	20.9	21.4
Wind output	P10	P10

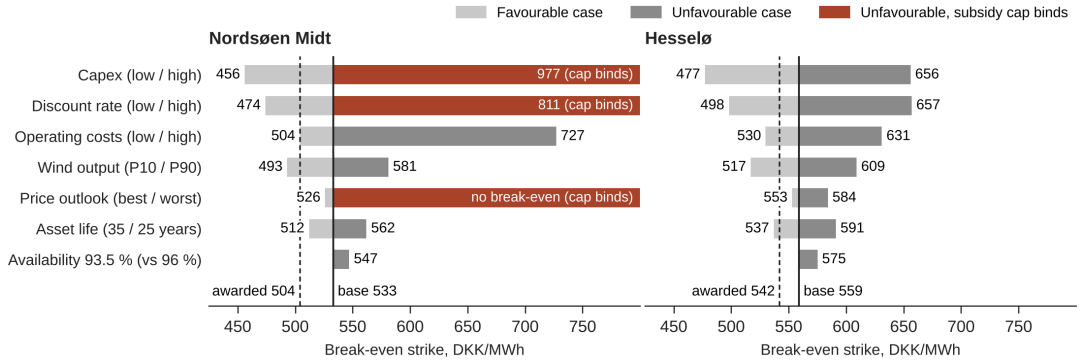
Base case: 5.0 % return (real, after tax), DKK 22.3m/MW (EUR 3.0m), P50 output. DKK 20.9m and 21.4m are EUR 2.8m and 2.9m. The levers are alternatives, not additive. P10 is a high-output case with a 10 % chance of being exceeded.

Key numbers

	Nordsøen Midt	Hesselø
Size and price zone	1,005 MW, West Denmark	795 MW, East Denmark
Awarded strike price	DKK 504/MWh (EUR 67.6)	DKK 542/MWh (EUR 72.7)
Break-even strike, base case	DKK 533/MWh	DKK 559/MWh
Project value (NPV) at the awarded price	−DKK 1.0bn (−EUR 0.13bn)	−DKK 0.45bn (−EUR 0.06bn)
State payment cap: average top-up it funds over 20 years	DKK 167/MWh	DKK 313/MWh
Return at the awarded price: 15 MW vs 20 MW turbines*	2.8 % vs 4.8 %	3.1 % vs 5.0 %
Vattenfall's discount rate for wind, real after tax†	3.6–5.3 % (all its wind assets)	

*The base case uses DEA costs for 2030 investment decisions, which assume 20 MW turbines, with output from a 15 MW turbine. The 15 MW case swaps in DEA costs for 15 MW units; the 20 MW case adds the matching 20 MW output. †Vattenfall Annual and Sustainability Report 2025, Note 27: 5.7–7.4 % after tax, read as nominal.

What moves the break-even price (one input at a time)



Red: the state's payment cap is used up, so a higher strike price adds almost no value and the break-even jumps or disappears. This happens three times at Nordsøen Midt and never at Hesselø.

What it means for the next tenders

Return requirements and turbine costs decided these bids more than wind did. Nordsøen Syd (bids due October 2028) has a DKK 6.6bn cap, under half of Nordsøen Midt's (about DKK 75/MWh of top-up), so its bidders must price low-price risk explicitly. (My interpretation.)

How it was built

Contract rules from the tender documents. Hourly wind 2006–2025 (ERA5) scaled to Energinet site measurements; 15 MW turbine. Danish prices 2016–2025 and the DEA's outlook. DEA costs adjusted for distance and depth. Monthly 30-year cash-flow model in Python with Danish tax rules; 45 tests.

Limits

Contract terms use the Nordsøen Syd documents as a proxy. The DEA's lower fleet price capture is run as a downside. Vattenfall's rate covers all its wind assets; tax pools follow one ruling (SKM2025.37.SR). Public data only; not affiliated with Vattenfall, the DEA or Energinet.

Skills: Python (pandas, SciPy), DCF and bid modelling, wind-resource analysis, tender-document review.

Code, data sources and tests: github.com/fergal567/danish-offshore-cfd-bid-model